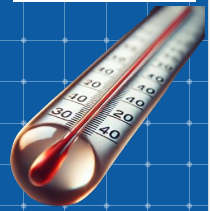
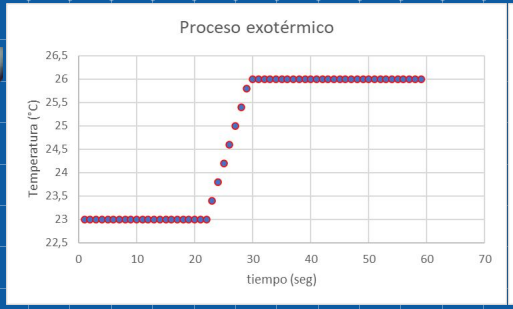
# Desarrollo en modalidad *open hardware* de un equipo calorimétrico.

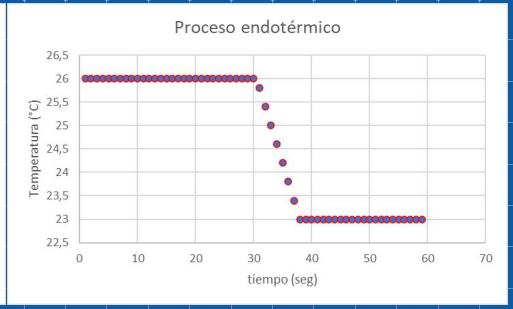
**Marcel Bentancor**
Especialización en Fabricación Digital e Innovación. UTEC-FabLab Barcelona

# 01

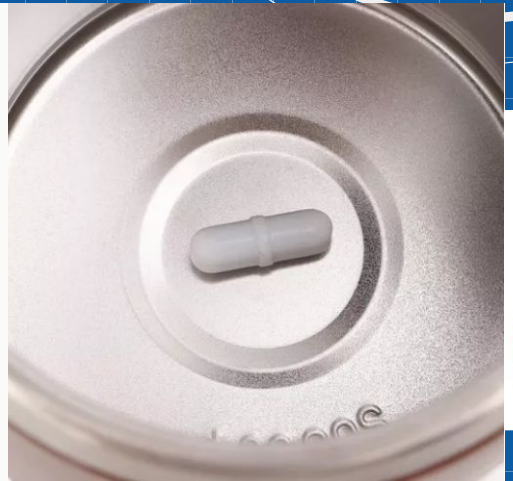
# ¿Qué hace?






# ¿Cómo llegaste a la idea?



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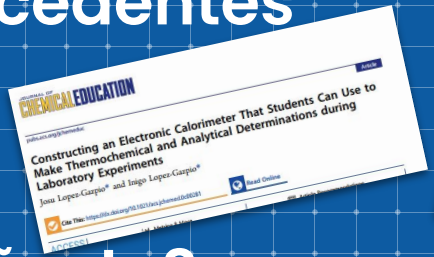
# 02 Antecedentes


JOURNAL OF CHEMICAL EDUCATION

Article

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Constructing an Electronic Calorimeter That Students Can Use to Make Thermochemical and Analytical Determinations during Laboratory Experiments

Jona Lopez-Garzón* and Iñigo Lopez-Garzón*

Cite This: https://dx.doi.org/10.1021/acs.jchemed.0c00281

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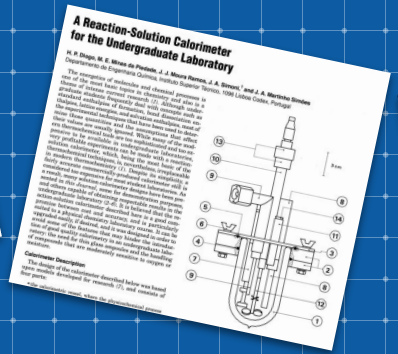
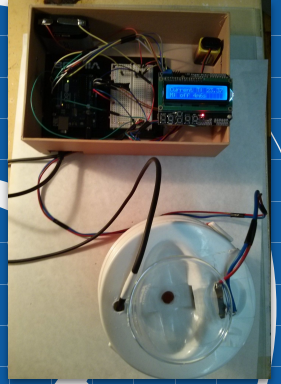

A Reaction-Solution Calorimeter for the Undergraduate Laboratory

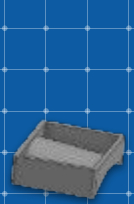
H. H. Diogo, M. E. Minas da Piedade, J. J. Moura Ramos, J. A. Simoni,* and J. A. Martinho Simões

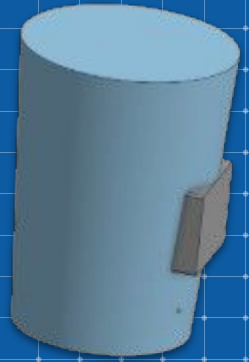
Departamento de Engenharia Química, Instituto Superior Técnico, 1096 Lisboa Codex, Portugal

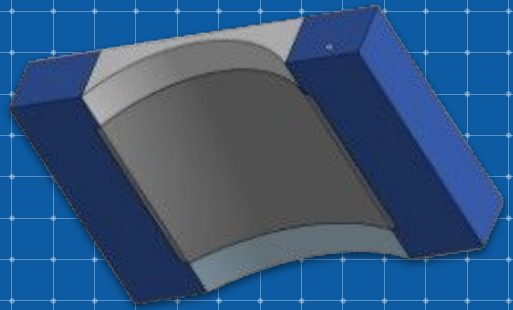
Calorimeter Description

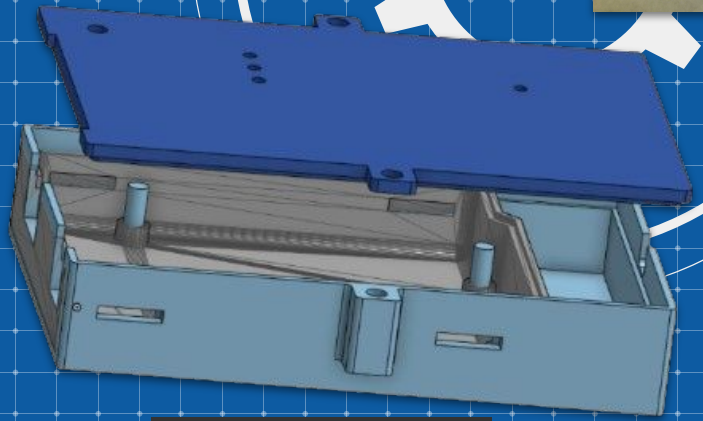


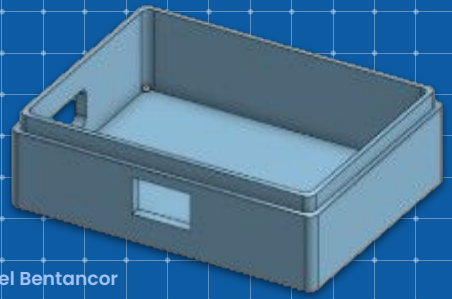
## ¿Qué diseñaste?

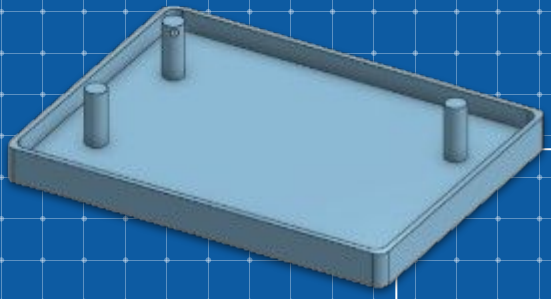


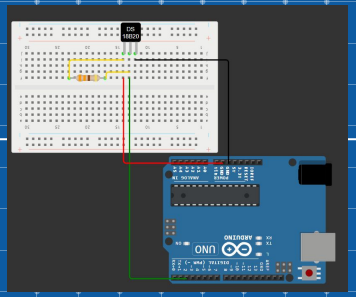










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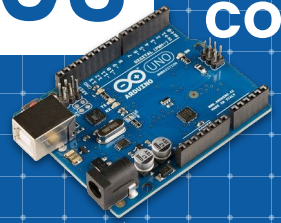
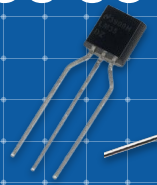
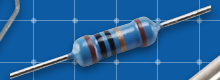
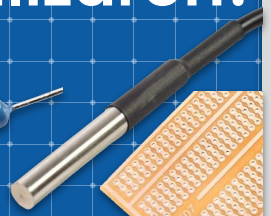
# 03

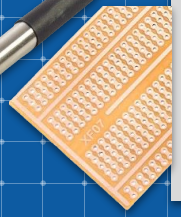
## ¿Qué materiales y componentes se utilizaron?

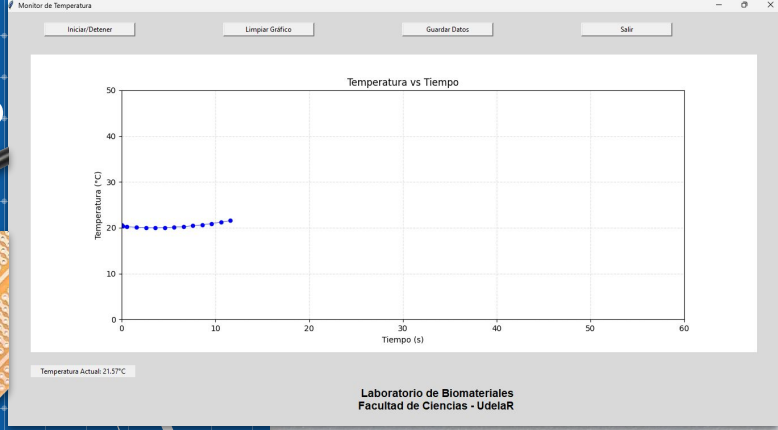




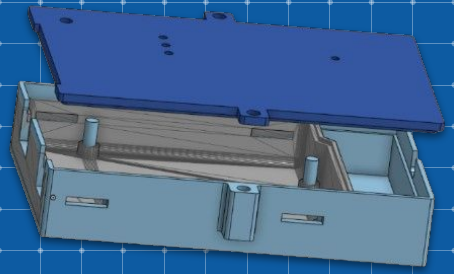


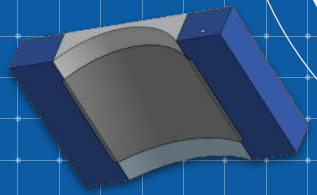




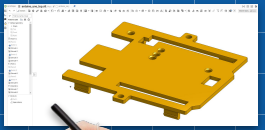


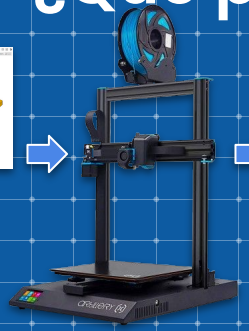

## ¿Qué partes y sistemas se fabricaron?

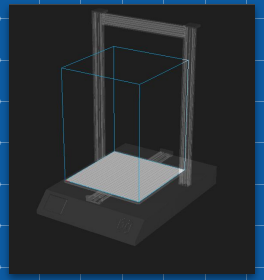


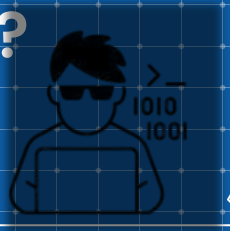


## ¿Qué procesos se utilizaron?









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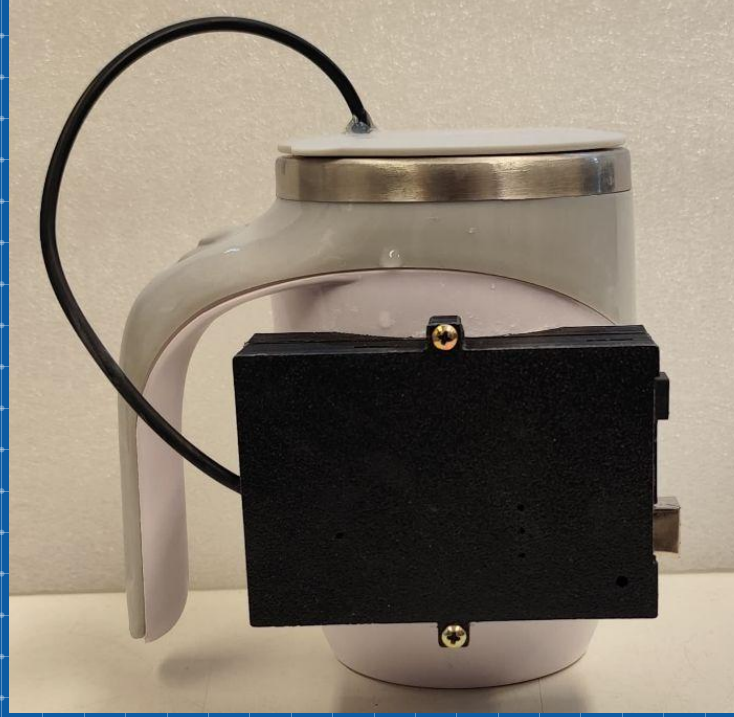
# 04 ¿Qué preguntas se respondieron?

¿Es posible construir un calorímetro de bajo costo, adaptable, y suficientemente robusto y apto para las clases prácticas?



## ¿Qué funcionó? ¿Qué no?

- ✅ Suficiente adiabaticidad
- ✅ Suficiente velocidad de respuesta
- ✅ Comunicación confiable
- ✅ Correcto registro en archivo .csv
- ✅ Interfaz gráfica funcional con zoom y desplazamiento
- ✅ Abscisas con tiempo transcurrido
- ✅ Dos versiones, con diferentes placas programables
- ✅ Resoluciones operativas diferentes

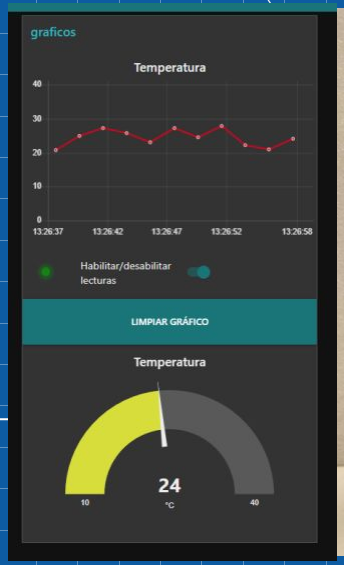
- 🚫 Interfaz gráfica estéticamente más llamativa

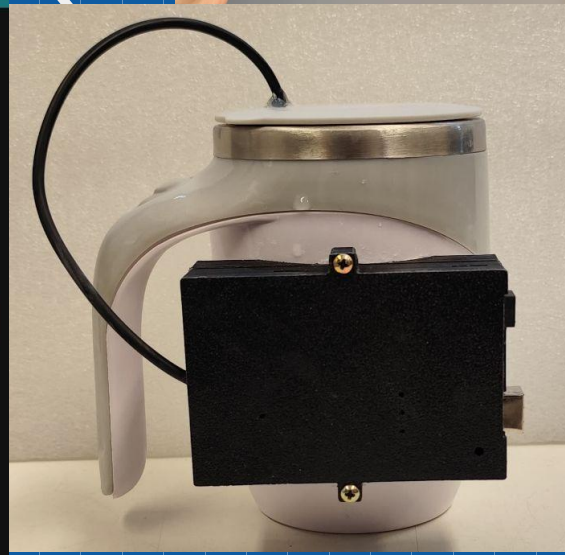
# 05 ¿Cuáles son las conclusiones?

Se logró diseñar y construir un calorímetro con registro automatizado de su temperatura interna, con alta velocidad de respuesta y suficiente exactitud para ser empleado en cursos prácticos universitarios sobre termoquímica o fisicoquímica general.

## ¿Cuáles son los pasos a seguir?

- Mejorar la estanqueidad del gabinete para facilitar su manipulación en clase.
- Se continuará la validación del calorímetro con otras reacciones químicas a fin de integrarlo a otras prácticas del curso.
- Se hará un seguimiento durante el primer año de uso en clases, para obtener retroalimentación de docentes y estudiantes.
- Se buscará mejorar la interfaz gráfica de usuario, para dotarla de un apariencia más atractiva y colorida.



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# Resumen gráfico

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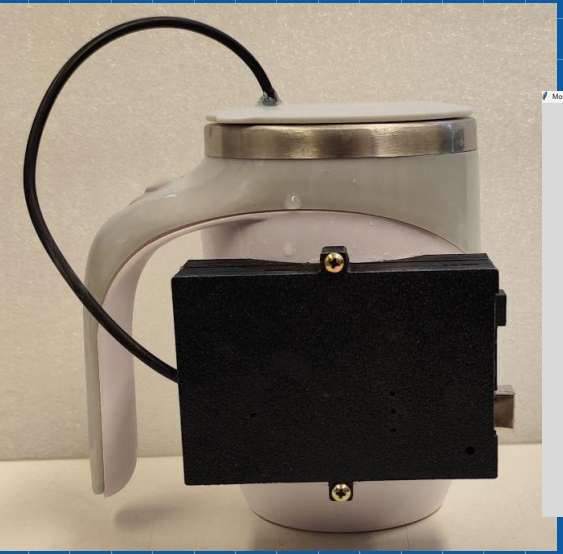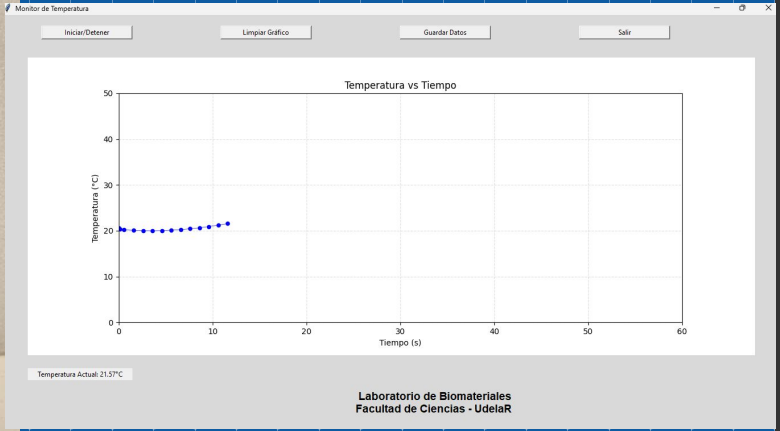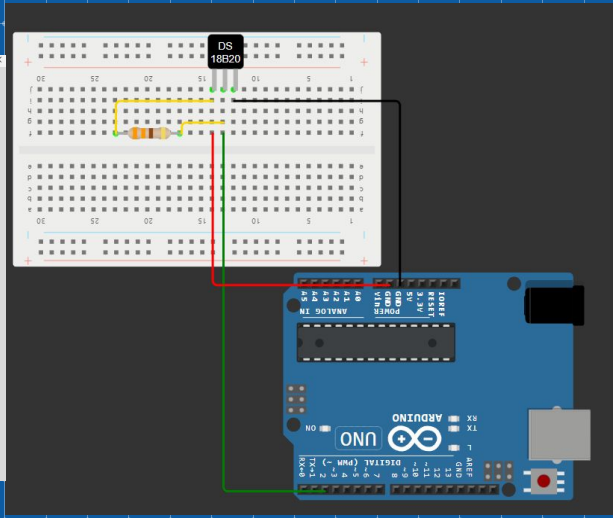
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